Copy_of_Students_performance_analysis.ipynb Save in GitHub to keep changes

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[ ]

```
%pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from panda
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (f
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pand
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dat
```

```python
import pandas as pd
from io import StringIO

csv_data = """
Name,Gender,Age,Math,Science,English,History,Attendance (%),Passed
Alice,Female,16,88,92,85,79,96,Yes
Bob,Male,17,75,78,72,70,88,Yes
Charlie,Male,16,65,60,58,62,85,No
Diana,Female,17,90,95,91,89,98,Yes
Ethan,Male,15,55,50,60,53,80,No
Fiona,Female,16,82,85,88,84,95,Yes
George,Male,17,70,65,68,60,87,No
Hannah,Female,15,95,98,93,90,99,Yes
Ian,Male,16,60,58,61,65,83,No
Julia,Female,17,85,87,84,82,94,Yes
"""

df = pd.read_csv(StringIO(csv_data))
display(df)
```

	Name	Gender	Age	Math	Science	English	History	Attendance (%)	Passed
0	Alice	Female	16	88	92	85	79	96	Yes
1	Bob	Male	17	75	78	72	70	88	Yes
2	Charlie	Male	16	65	60	58	62	85	No
3	Diana	Female	17	90	95	91	89	98	Yes
4	Ethan	Male	15	55	50	60	53	80	No
5	Fiona	Female	16	82	85	88	84	95	Yes
6	George	Male	17	70	65	68	60	87	No
7	Hannah	Female	15	95	98	93	90	99	Yes
8	Ian	Male	16	60	58	61	65	83	No
9	Julia	Female	17	85	87	84	82	94	Yes

Start coding or generate with AI.

[ ]

```
ydf.head(5)
```

	Name	Gender	Age	Math	Science	English	History	Attendance (%)	Passed
0	Alice	Female	16	88	92	85	79	96	Yes
1	Bob	Male	17	75	78	72	70	88	Yes
2	Charlie	Male	16	65	60	58	62	85	No
3	Diana	Female	17	90	95	91	89	98	Yes
4	Ethan	Male	15	55	50	60	53	80	No

```
[ ]
df.tail(6)
```

	Name	Gender	Age	Math	Science	English	History	Attendance (%)	Passed
4	Ethan	Male	15	55	50	60	53	80	No
5	Fiona	Female	16	82	85	88	84	95	Yes
6	George	Male	17	70	65	68	60	87	No
7	Hannah	Female	15	95	98	93	90	99	Yes
8	Ian	Male	16	60	58	61	65	83	No
9	Julia	Female	17	85	87	84	82	94	Yes

[ ]

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 9 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Name            10 non-null     object
 1   Gender          10 non-null     object
 2   Age             10 non-null     int64
 3   Math            10 non-null     int64
 4   Science         10 non-null     int64
 5   English         10 non-null     int64
 6   History         10 non-null     int64
 7   Attendance (%)  10 non-null     int64
 8   Passed          10 non-null     object
dtypes: int64(6), object(3)
memory usage: 852.0+ bytes
```

[ ]

```
df.dtypes
```

	0
Name	object
Gender	object
Age	int64
Math	int64
Science	int64
English	int64
History	int64
Attendance (%)	int64
Passed	object

dtype: object

```
df.describe()
```

	Age	Math	Science	English	History	Attendance (%)
count	10.000000	10.000000	10.00000	10.000000	10.000000	10.000000
mean	16.200000	76.500000	76.80000	76.000000	73.400000	90.500000
std	0.788811	13.623917	17.24851	13.695092	13.099618	6.720615
min	15.000000	55.000000	50.00000	58.000000	53.000000	80.000000
25%	16.000000	66.250000	61.25000	62.750000	62.750000	85.500000
50%	16.000000	78.500000	81.50000	78.000000	74.500000	91.000000
75%	17.000000	87.250000	90.75000	87.250000	83.500000	95.750000
max	17.000000	95.000000	98.00000	93.000000	90.000000	99.000000

```
[ ]
df.shape

(10, 9)
```

```
[ ]
df.columns

Index(['Name', 'Gender', 'Age', 'Math', 'Science', 'English', 'History',
       'Attendance (%)', 'Passed'],
      dtype='object')
```

```
[ ]
df.values

array([['Alice', 'Female', 16, 88, 92, 85, 79, 96, 'Yes'],
       ['Bob', 'Male', 17, 75, 78, 72, 70, 88, 'Yes'],
       ['Charlie', 'Male', 16, 65, 60, 58, 62, 85, 'No'],
       ['Diana', 'Female', 17, 90, 95, 91, 89, 98, 'Yes'],
       ['Ethan', 'Male', 15, 55, 50, 60, 53, 80, 'No'],
       ['Fiona', 'Female', 16, 82, 85, 88, 84, 95, 'Yes'],
       ['George', 'Male', 17, 70, 65, 68, 60, 87, 'No'],
       ['Hannah', 'Female', 15, 95, 98, 93, 90, 99, 'Yes'],
       ['Ian', 'Male', 16, 60, 58, 61, 65, 83, 'No'],
       ['Julia', 'Female', 17, 85, 87, 84, 82, 94, 'Yes']], dtype=object)
```

```
filtered_df=df.query('English>50')
display(filtered_df)
```

	Name	Gender	Age	Math	Science	English	History	Attendance (%)	Passed
0	Alice	Female	16	88	92	85	79	96	Yes
1	Bob	Male	17	75	78	72	70	88	Yes
2	Charlie	Male	16	65	60	58	62	85	No
3	Diana	Female	17	90	95	91	89	98	Yes
4	Ethan	Male	15	55	50	60	53	80	No
5	Fiona	Female	16	82	85	88	84	95	Yes
6	George	Male	17	70	65	68	60	87	No
7	Hannah	Female	15	95	98	93	90	99	Yes
8	Ian	Male	16	60	58	61	65	83	No
9	Julia	Female	17	85	87	84	82	94	Yes

```
df_droppedI =df.drop(columns=['Age'])
display(df_droppedI)
```

	Name	Gender	Math	Science	English	History	Attendance (%)	Passed
0	Alice	Female	88	92	85	79	96	Yes
1	Bob	Male	75	78	72	70	88	Yes
2	Charlie	Male	65	60	58	62	85	No
3	Diana	Female	90	95	91	89	98	Yes
4	Ethan	Male	55	50	60	53	80	No
5	Fiona	Female	82	85	88	84	95	Yes
6	George	Male	70	65	68	60	87	No
7	Hannah	Female	95	98	93	90	99	Yes
8	Ian	Male	60	58	61	65	83	No
9	Julia	Female	85	87	84	82	94	Yes

```
[ ]
df[['Math', 'Science', 'English', 'History']].mean()
```

	0
Math	76.5
Science	76.8
English	76.0
History	73.4

dtype: float64

```
[ ]
passed_students = df[df['Passed'] == 'Yes']
failed_students = df[df['Passed'] == 'No']
```

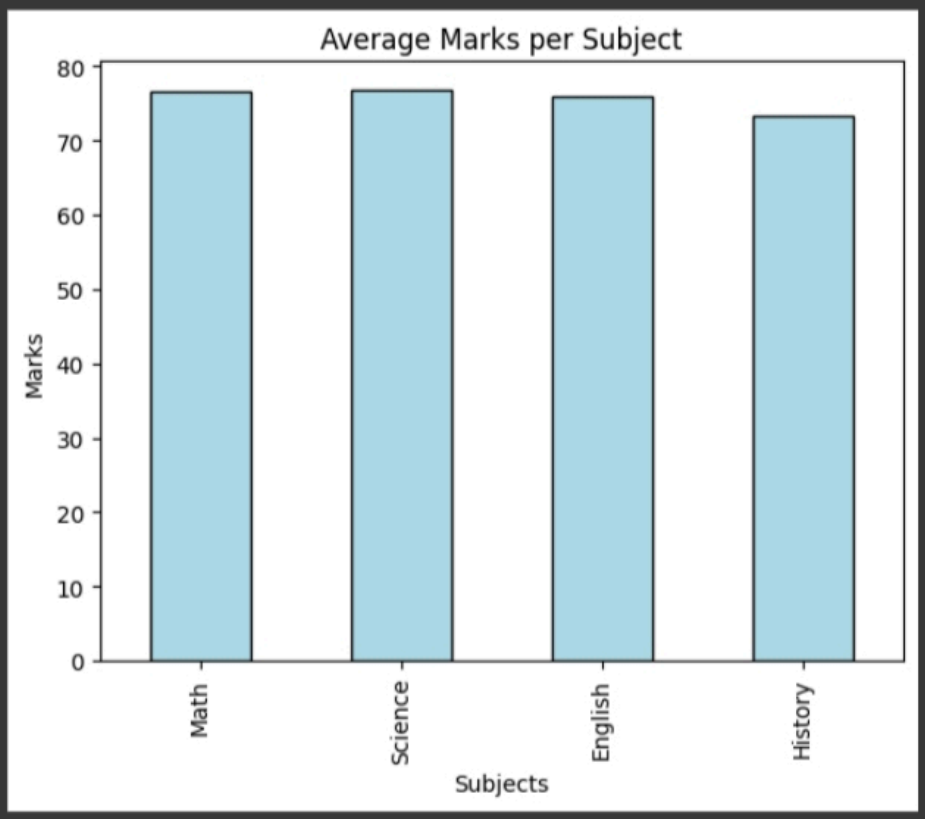
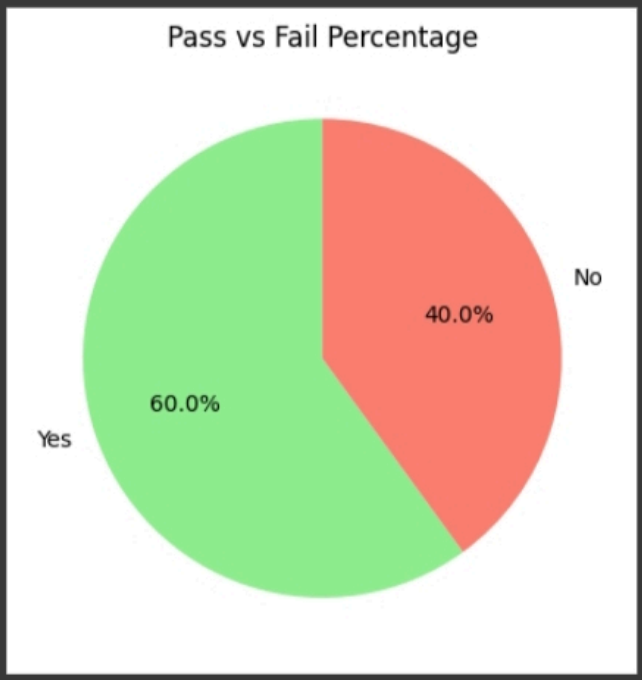
```
[ ]
print("\nCorrelation between Attendance and Marks:\n")
print(df[['Attendance (%)', 'Math', 'Science', 'English', 'History']].corr())
```

```
Correlation between Attendance and Marks:

                Attendance (%)      Math   Science   English   History
Attendance (%)        1.000000  0.989625  0.983432  0.973013  0.961711
Math                  0.989625  1.000000  0.992468  0.959370  0.942590
Science               0.983432  0.992468  1.000000  0.965673  0.956854
English               0.973013  0.959370  0.965673  1.000000  0.954413
History               0.961711  0.942590  0.956854  0.954413  1.000000
```

```
[ ]
import matplotlib.pyplot as plt

subject_avg = df[['Math', 'Science', 'English', 'History']].mean()
subject_avg.plot(kind='bar', color='lightblue', edgecolor='black')
plt.title('Average Marks per Subject')
plt.ylabel('Marks')
plt.xlabel('Subjects')
plt.show()
```



```
df['Passed'].value_counts().plot(kind='pie', autopct='%1.1f%%', startangle=90, colors=['lightgreen
plt.title('Pass vs Fail Percentage')
plt.ylabel('')
plt.show()
```